QSS 20 Final Project Milestone 1

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1 Project Option

Senior Thesis / Own Data. I am continuing the area I committed to in *Me Search* and developed in my literature review: why children born into similar families end up with different adult outcomes depending on where they grow up. My data are the tract-level files from Opportunity Insights' Opportunity Atlas (`https://opportunityinsights.org/data/`).

2 Your project's main questions

My literature review sharpened my original question about the rich–poor achievement gap into: **what distinguishes rural Census tracts with above-average upward mobility from otherwise similar rural tracts with below-average mobility?** The existing literature is overwhelmingly urban (Moving to Opportunity, the neighborhood-exposure papers, and the segregation literature all take the metropolitan neighborhood as the unit of interest), and the Atlas paper itself notes that "rurality" helps in some regions and hurts in others without saying why. So, I want to know which neighborhood characteristics separate high- from low-mobility rural tracts, and whether those correlates differ from the ones that predict mobility in dense urban tracts.

3 Your project's data sources/relevant fields

I inner-joined `tract_outcomes_simple.csv` (73,278 tracts) and `tract_covariates.csv` (74,044 tracts) on `state`, `county`, and `tract`, giving 73,199 matches and 71,958 tracts after dropping those missing mobility or density. The unit of analysis is the 2010 Census tract, defined by where a child *grew up*; the children are the 1978–1983 birth cohorts, with adult outcomes measured around ages 31–37 from tax records. My outcome is `kfr_pooled_pooled_p25`, the mean adult household income *rank* for children whose parents sat at the 25th percentile. Candidate explanatory fields include `popdensity2000` (binned into density categories as my rural measure), `singleparent_share2000`, `poor_share2000`, `frac_coll_plus2000`, `emp2000`, `share_white2000`, `gsmn_math_g3_2013`, and `ann_avg_job_growth_2004_2013`. Every outcome ships with a standard error and a cell count, which I plan to use to weight or screen out tracts resting on very few children.

4 Create one starter visualization

Among the 17,978 rural tracts (under 100 people per square mile), I split at the national tract median of mobility (9,087 land above it) and compared the two groups on ten characteristics, standardized so different scales are comparable. Single-parent household share is the biggest separator at −0.81 SD, followed by poverty at −0.65 SD. Job growth, which I expected to matter for rural economies, is nearly flat at +0.09 SD, consistent with the literature's claim that mobility is a childhood-environment story rather than a labor-demand one. Share white is the largest positive gap, which I will have to handle carefully rather than report as a finding; see Section 6.

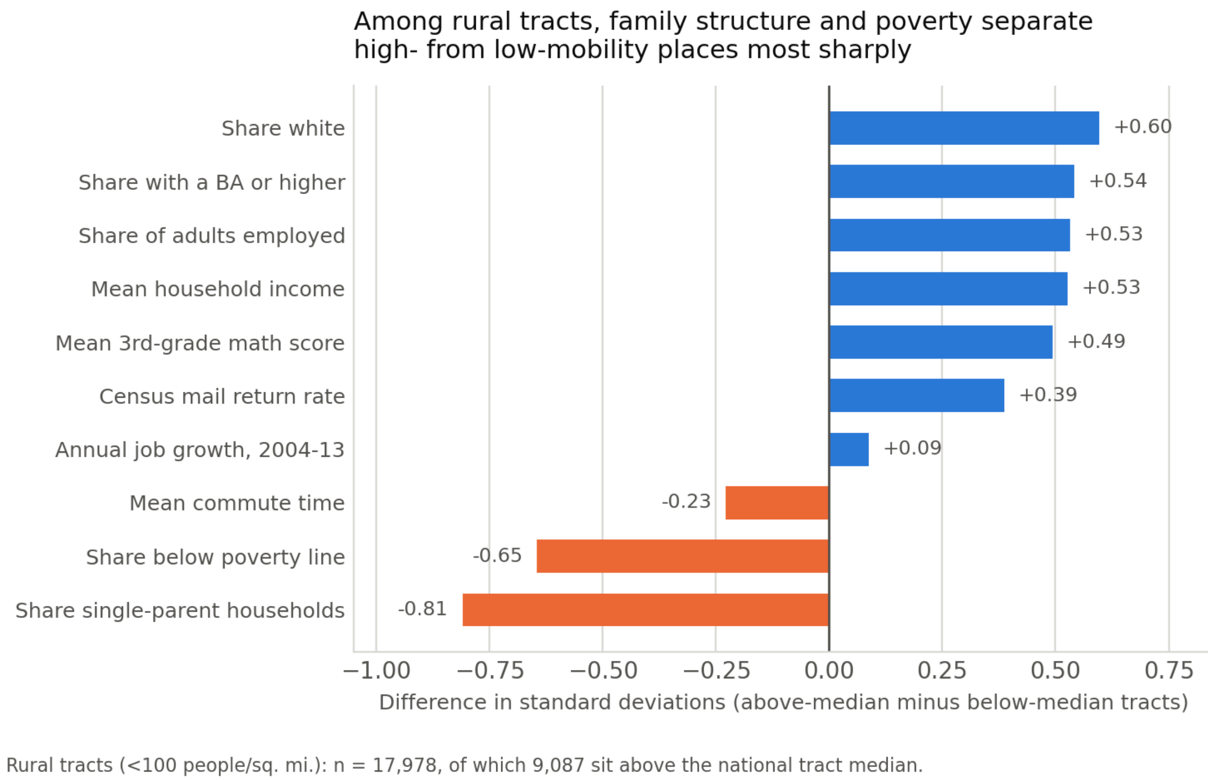

Figure 1: Standardized difference in neighborhood characteristics between rural tracts with above-median and below-median upward mobility. Bars show the difference in group means divided by the standard deviation of that characteristic among rural tracts; blue bars are higher in high-mobility tracts, orange bars are lower. Differences are unadjusted: these are correlations, not effects.

5 Learn from an earlier project on your research topic

I read the Fall 2022 descriptive-comparison example, "Exploring the Association Between Race and Bail Decisions in Cook County, IL" (Fink, Jin, Punaini, and Tolat).

Lesson. They refused to compare pooled averages. Rather than reporting one Black–White gap, they compared rates *within* each of the five most common offenses, because a pooled gap would mostly reflect which offenses each group was charged with. They also named the confounder they could not observe, prior criminal history. My figure is currently a pooled comparison of exactly the kind they avoided, since high- and low-mobility rural tracts also differ in region and racial composition at once. Before writing anything up as a finding I will stratify within commuting zone and within bands of racial composition.

Weakness I will improve on. They report point estimates almost entirely without uncertainty: only Table 2 carries p-values, while Tables 3–8 give bare differences with no standard errors, intervals, or sample sizes. This becomes a real problem in their Table 8, where ratios computed on base rates as small as 0.000098 support a claim that Black defendants are "as much as 700% more likely" to see a bond increase. Small rural tracts are exactly where the Atlas standard errors are largest, so I will attach intervals to every difference I report, precision-weight or impose a minimum-cell-count screen, and show how conclusions shift across screens.

6 Your project's anticipated challenges

The most likely way this project fails is that "rural" turns out to be a proxy for "white and non-poor" and I write up a race gap as a rurality finding; the figure already shows that warning sign, and since high-mobility rural tracts cluster in places like the upper Midwest, any comparison risks picking up regional composition instead. The fix is within-region comparisons plus reporting mobility separately by child race, which the Atlas supports. Measurement is another issue. Density is not a great stand in for rurality, my 100-per-square-mile cutoff is my choice rather than a standard, and several covariates are measured after the cohorts I study grew up. Third, missingness is not random: the sparsest rural tracts are the most likely to be suppressed or noisy, so dropping them quietly changes what "rural" means in my sample. Fourth, all of this is cross-sectional correlation across places and cannot separate a neighborhood effect from families sorting into neighborhoods. I plan to scope the project as explicitly descriptive and keep causal language out unless I can find a design that supports it.